

Varun Kumar Atkuri

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PROFESSIONAL SUMMARY:

Results-oriented Data Engineer with 5+ years of experience building scalable, cloud-native data pipelines across **Healthcare** and **Retail** domains. Proficient in the modern data stack, specializing in **Snowflake (Snowpark)**, **dbt**, **Python**, and the complete **AWS Ecosystem** (S3, EMR, Glue, Kinesis). Hands-on experience in developing modular data models, implementing incremental **ELT workflows**, and supporting analytics and **Machine Learning** teams with high-quality, **HIPAA-compliant** datasets. Strong focus on pipeline performance, cost optimization, and data governance best practices in Agile environments.

PROFESSIONAL EXPERIENCE:

McKesson | Data Engineer | Atlanta, GA (Remote) | Aug 2024 – Present

Project: Enterprise Healthcare Data Lake and Analytics Platform (AWS Cloud)

- **Architected and built layered dbt models** in Snowflake, designing modular data marts (Facts/Dimensions) to support high-performance analytics and downstream ML pipelines.
- **Engineered point-in-time feature tables** (using "as-of" joins) to ensure historical data accuracy for the Data Science team's disease prediction models.
- **Implemented incremental dbt strategies** for TB-scale fact tables, utilizing surrogate keys and watermark columns to reduce daily compute costs by **18%**.
- **Built robust ingestion pipelines** using **Snowpipe**, automatically loading raw JSON/CSV data from **Amazon S3** into Snowflake staging tables with near real-time latency.
- **Extended dbt transformations** with **Snowpark Python UDFs** to parse complex HL7 and ICD/CPT codes, enabling efficient in-warehouse data cleaning.
- **Orchestrated end-to-end data workflows** using **Apache Airflow**, using sensors to validate **data availability** before triggering dbt transformations to meet the **15-minute** freshness SLA.
- **Established CI/CD pipelines** using **GitHub Actions**, automating SQL linting (SQLFluff), dbt model testing, and production deployments to ensure code quality.
- **Automated data integration with AWS SageMaker**, exporting cleaned datasets to S3 for model training and ingesting inference results back into Snowflake.
- **Designed and deployed interactive Tableau dashboards** connected to Snowflake Data Marts, visualizing patient risk scores and operational KPIs to enable clinical decision-making.
- **Implemented Data Quality (DQ) safeguards** using **dbt tests** (unique, not_null) and custom SQL checks, creating quarantine flows that reduced data defects by **12%**.
- **Managed Infrastructure as Code (IaC)** using **Terraform** to provision AWS resources (S3, IAM, Glue) and Snowflake objects, ensuring environment consistency.
- **Applied HIPAA-aligned security** controls, implementing Snowflake **RBAC**, Dynamic Masking policies, and AWS KMS encryption to protect sensitive PII/PHI data.
- **Optimized Snowflake performance** by configuring **Clustering Keys** and analyzing Query Profiles, improving retrieval speeds on large transaction tables by **40%**.
- **Configured Observability** using **Amazon CloudWatch** and Snowflake Resource Monitors to track pipeline failures and credit usage, preventing budget overruns.

Tech Stack: Python, SQL, dbt, Snowflake (Snowpipe, Snowpark, Tasks), AWS (S3, Glue, IAM, KMS, CloudWatch), Apache Airflow, GitHub Actions, Terraform, Tableau, Git.

Nisum Technologies | Data Engineer | Hyderabad, India | Nov 2021 – Aug 2023

Project: Medical Claims Processing & Analytics Platform (AWS)

- **Developed and maintained** scalable batch ETL pipelines for processing high-volume Medical Claims data, collaborating with architects to ensure accurate financial reporting.
- **Built raw data ingestion workflows**, loading daily claims dumps and member eligibility files from SFTP servers into **Amazon S3** (Bronze Layer) using Python scripts and **AWS Lambda** triggers.
- **Wrote complex PySpark transformations on Amazon EMR** to validate claim headers, standardize provider NPIs, and compute "allowed vs. billed" amounts, processing **20M+ records per batch**.
- **Migrated legacy SQL logic to Spark SQL on EMR**, transforming raw data into optimized **Parquet format** (Silver Layer) on S3 to handle schema evolution and improve query performance.
- **Contributed to real-time data streaming** using **Amazon Kinesis Data Streams**, capturing claim status changes (Approved/Denied/Pending) to feed live operational dashboards.
- **Orchestrated end-to-end data workflows** using **AWS Step Functions**, coordinating dependencies between EMR Spark jobs, Glue Crawlers, and Redshift loads with automated retry logic.
- **Optimized EMR PySpark jobs** by implementing partitioning by Claim_Date and Service_Type, and broadcasting lookup tables (Member/Provider), reducing job runtime by **25%**.
- **Developed Redshift Data Marts** (Gold Layer) for the finance team, designing tables with appropriate **Distribution and Sort Keys** to speed up queries on aggregate claim costs.
- **Implemented Data Quality checks** in the pipeline (using PySpark) to validate Nulls/Duplicates, effectively reducing data defects in the final Redshift reports by **25%**.
- **Supported the DevOps culture** by deploying AWS infrastructure (S3 buckets, IAM Roles, EMR Security Configs) using **Terraform** and managing code versions via **Git/GitHub Actions**.
- **Monitored production pipelines** using **Amazon CloudWatch**, creating alarms for job latencies and failures, and performed root cause analysis (RCA) to resolve P1/P2 data incidents.
- **Assisted in cataloging data assets** using **AWS Glue Data Catalog**, ensuring all tables in the Data Lake were discoverable for downstream analytics users in Athena.
- **Collaborated on cost-optimization tasks**, configuring **EMR Managed Scaling** policies to scale down worker nodes during off-peak hours, saving **15%** on compute costs.
- **Enforced security standards** by applying IAM policies and S3 Bucket Policies, ensuring strict access controls for PII/PHI data in compliance with HIPAA regulations.

Tech Stack: Python, SQL, AWS EMR (PySpark), S3, Kinesis, Redshift, Athena, Glue, Step Functions, Terraform, Docker, CloudWatch, IAM, Lake Formation, Tableau, Git.

Excers.com | Data Engineer | Hyderabad, India | Mar 2020 – Oct 2021

Project: Retail Hadoop to AWS Redshift Migration

- **Executed data migration scripts** to move 1.5 TB/day of historical data from legacy Hive tables to **Amazon Redshift**, achieving 99.98% row-level accuracy.
- **Developed Spark jobs** to convert raw CSV files into partitioned **Parquet formats** in S3, enabling faster downstream querying and secure storage using SSE-KMS.
- **Implemented incremental load logic** using staging tables to handle daily data updates, which helped reduce the ETL window by 22%.
- **Monitored and tuned Redshift performance** by running **VACUUM** and **ANALYZE** commands and applying recommended SORT/DIST keys to improve query speed.
- **Built data validation scripts** (row counts, null checks) to identify data discrepancies between the old Hadoop system and the new Cloud warehouse.
- **Automated bad-data handling** by routing failed records to a **quarantine S3 bucket** for manual review, reducing the need for full pipeline re-runs.
- **Scheduled and monitored ETL DAGs** in **Apache Airflow**, configuring parameterized variables to easily rerun jobs for different dates.
- **Created CloudWatch alerts** to notify the team of pipeline failures and built basic **Grafana dashboards** to track

daily load completion times.

- **Performed UAT data reconciliation**, generating reports on key metrics (KPIs) to support the senior team in obtaining migration sign-off.
- **Maintained documentation** for ETL mappings and runbooks, ensuring smooth knowledge transfer for operational support teams.

***Tech Stack:** Python, SQL, Apache Spark, Hive, Amazon S3, Redshift, Parquet, Airflow, CloudWatch, Grafana, Git/Bitbucket.*

EDUCATION:

- **University of North Texas, Denton, TX** — M.S. Data Science, **GPA 3.9** (Aug 2023 – May 2025)
- **Kalasalingam University, Tamil Nadu, India** — B.S. Mechanical Engineering, **GPA 3.3** (Jun 2017 – Mar 2021)

CERTIFICATIONS:

- AWS Certified Data Engineer – Associate
- AWS Certified Machine Learning Engineer – Associate
- AWS Certified Cloud Practitioner
- SnowPro Core Certified
- Fundamentals of the Databricks Generative AI
- Prompt Engineering for ChatGPT by Vanderbilt University (Coursera)

TECHNICAL SKILLS:

Programming & Scripting	Python, SQL, Java, Scala, Bash/Shell, Linux
Data Engineering & ETL	Apache Airflow, AWS Glue, Azure Data Factory (ADF), dbt, Informatica, Talend, Apache NiFi, SSIS, Fivetran
Cloud Platforms & Storage	Amazon Web Services (AWS), Microsoft Azure, Google Cloud Platform (GCP)
Data Warehousing & Lakes	Snowflake, Amazon Redshift, Google BigQuery, Hadoop (HDFS)
Big Data & Stream Processing	Apache Spark (PySpark), Apache Hadoop, Azure Databricks, Apache Kafka, Apache Flume, Google Pub/Sub
Databases (SQL & NoSQL)	Oracle, MySQL, PostgreSQL, MongoDB, Cassandra, Spark SQL
BI & Visualization	Tableau, Power BI, SSRS
Monitoring & Observability	Grafana, CloudWatch, ELK Stack , Dynatrace, Quantum Metric
DevOps & CI/CD	Git, GitHub, Bitbucket, Jenkins, Docker, Kubernetes, Terraform, AWS CloudFormation
Data Governance & Quality	dbt Tests, AWS Deequ, Glue Data Quality, RBAC, Masking Policies
MLOps & Feature Engineering	Snowpark, SageMaker, MLflow, Feature Store Concepts, Model Monitoring
IDEs & Notebooks	Jupyter Notebook, IntelliJ IDEA, Visual Studio Code, Google Colab
Machine Learning & AI	Scikit-learn, spaCy, NLTK, Basic Deep Learning, LLM Concepts, Prompt Engineering